

MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING & INFORMATICS

DATABASE SYSTEMS

CSC1204

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REAL ESTATE MANAGEMENT SYSTEM REPORT

Executive Summary

The Real Estate Management System is a web-based database application developed to simplify and improve the management of real estate activities such as client registration, plot management, viewing scheduling, and reservation processing. The project was designed using PHP, MySQL, HTML, and CSS. The purpose of the system is to reduce manual paperwork, improve data accuracy, and provide an organized method of handling real estate transactions.

The system enables real estate agents to register clients, manage plot availability, schedule property viewings, and reserve plots for interested customers. The project integrates database management principles with a user-friendly interface to provide an efficient and reliable solution for property management operations.

Business Case

Introduction

Real estate companies handle large amounts of information involving clients, plots, reservations, and property viewings. In many small organizations, these activities are still managed manually using books, paper files, and spreadsheets. Such methods are slow, inefficient, and highly vulnerable to errors, data loss, duplication, and poor record management.

The Real Estate Management System was developed to solve these challenges by providing an automated database-driven application that stores and manages information efficiently. The

system improves communication between agents and clients while ensuring that property information is updated in real time.

Problem Statement

Many real estate organizations face several operational problems including:

1. Difficulty tracking available and reserved plots.
2. Poor organization of client information.
3. Delays in scheduling and managing property viewings.
4. Manual reservation processes that lead to errors.
5. Risk of losing important records.
6. Lack of a centralized system for property management.

These challenges reduce efficiency and negatively affect customer service.

Objectives of the Project

The main objective of the project is to develop a database-driven web application that automates the management of real estate activities.

Specific Objectives

1. To create a system for registering and managing clients.
2. To manage plot information and plot status.
3. To schedule property viewings efficiently.
4. To allow reservation of plots with deposit recording.
5. To provide a user-friendly web interface.
6. To improve data storage, retrieval, and security.

Scope of the Project

The project focuses on managing the following operations:

- Client registration
- Plot management

- Viewing scheduling
- Reservation management
- Plot status updates
- Database storage and retrieval.

Benefits of the System

The Real Estate Management System provides several benefits including:

- Faster processing of client and reservation information.
- Reduced paperwork and manual errors.
- Improved organization of records.
- Better customer service.
- Easy access to stored information.
- Improved decision making through organized data.

Design

Logical Design

The logical design explains the structure of the database and the relationships between entities in the system.

Entity Description

Users Table

The users table stores information

```
MariaDB [real_estate]> DESCRIBE users;
```

Field	Type	Null	Key	Default	Extra
user_id	int(11)	NO	PRI	NULL	auto_increment
username	varchar(50)	NO		NULL	
role	enum('admin','agent')	YES		agent	
created_at	timestamp	NO		current_timestamp()	

```
4 rows in set (0.069 sec)
```

```
MariaDB [real_estate]> SELECT*FROM users;
```

user_id	username	role	created_at
1	Admin Agent	agent	2026-05-05 20:53:28

```
1 row in set (0.001 sec)
```

Plots Table

The plots table stores information about available, reserved, and sold plots.

```
MariaDB [real_estate]> DESCRIBE plots;
```

Field	Type	Null	Key	Default	Extra
plot_id	int(11)	NO	PRI	NULL	auto_increment
plot_number	varchar(50)	NO		NULL	
location	varchar(100)	YES		NULL	
price	decimal(15,2)	YES		NULL	
status	enum('available','reserved','sold')	YES		available	

```
5 rows in set (0.062 sec)
```

```
MariaDB [real_estate]> SELECT*FROM plots;
```

plot_id	plot_number	location	price	status
1	P001	Kampala - Ntinda	25000000.00	reserved
2	P002	Kampala - Kira	18000000.00	reserved
3	P003	Wakiso - Entebbe Road	35000000.00	available
4	P004	Mukono Town	12000000.00	available
5	P005	Jinja - Bugembe	8000000.00	available

Clients Table

The clients table stores customer information.

```
MariaDB [real_estate]> DESCRIBE clients;
```

Field	Type	Null	Key	Default	Extra
client_id	int(11)	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
phone	varchar(20)	YES		NULL	
email	varchar(100)	YES		NULL	
interest_type	varchar(100)	YES		NULL	
date_registered	timestamp	NO		current_timestamp()	

6 rows in set (0.086 sec)

```
MariaDB [real_estate]> SELECT*FROM clients;
```

client_id	name	phone	email	interest_type	date_registered
1	KANGWAGYE JONAS	+256765304174	NULL		2026-05-05 20:35:26
2	KANGWAGYE JONAS	+256765304174	NULL	residential	2026-05-05 20:55:43

2 rows in set (0.008 sec)

Viewing Table

The viewing table stores details about scheduled property viewings.

```
MariaDB [real_estate]> DESCRIBE viewing;
```

Field	Type	Null	Key	Default	Extra
viewing_id	int(11)	NO	PRI	NULL	auto_increment
plot_id	int(11)	NO	MUL	NULL	
client_id	int(11)	NO	MUL	NULL	
agent_id	int(11)	NO	MUL	NULL	
viewing_date	date	YES		NULL	
viewing_time	time	YES		NULL	
status	enum('scheduled','completed','cancelled')	YES		scheduled	

rows in set (0.063 sec)

```
MariaDB [real_estate]> SELECT*FROM viewing;
```

viewing_id	plot_id	client_id	agent_id	viewing_date	viewing_time	status
5	4	1	1	2026-05-26	NULL	scheduled

row in set (0.002 sec)

Reservation Table

The reservation table stores plot reservation information.

```
MariaDB [real_estate]> DESCRIBE reservation;
```

Field	Type	Null	Key	Default	Extra
reservation_id	int(11)	NO	PRI	NULL	auto_increment
plot_id	int(11)	NO	MUL	NULL	
client_id	int(11)	NO	MUL	NULL	
agent_id	int(11)	NO	MUL	NULL	
deposit_amount	decimal(15,2)	YES		NULL	
status	enum('reserved','not reserved')	YES		reserved	
reservation_date	timestamp	NO		current_timestamp()	

```
7 rows in set (0.052 sec)
```

```
MariaDB [real_estate]> SELECT*FROM reservation;
```

reservation_id	plot_id	client_id	agent_id	deposit_amount	status	reservation_date
2	2	1	1	500000.00	reserved	2026-05-05 20:58:56
3	1	1	1	5000000000.00	reserved	2026-05-05 20:59:37
4	1	1	1	5000000000.00	reserved	2026-05-05 21:02:31

```
3 rows in set (0.001 sec)
```

NOTIFICATIONS

```
MariaDB [real_estate]> DESCRIBE notifications;
```

Field	Type	Null	Key	Default	Extra
notification_id	int(11)	NO	PRI	NULL	auto_increment
user_id	int(11)	NO	MUL	NULL	
message	text	YES		NULL	
created_at	timestamp	NO		current_timestamp()	

```
4 rows in set (0.058 sec)
```

```
MariaDB [real_estate]> SELECT*FROM notifications;
```

notification_id	user_id	message	created_at
1	1	Plot P001 has been reserved with deposit of 5000000000	2026-05-05 21:02:31
2	1	New feedback submitted for viewing #5 - Interest: high	2026-05-05 21:03:44

```
2 rows in set (0.001 sec)
```

FEEDBACK

```

MariaDB [real_estate]> DESCRIBE feedback;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| feedback_id    | int(11)       | NO   | PRI | NULL    | auto_increment |
| viewing_id     | int(11)       | NO   | MUL | NULL    |                |
| comments       | text          | YES  |     | NULL    |                |
| interest_level | enum('high','medium','low') | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.049 sec)

MariaDB [real_estate]> SELECT*FROM feedback;
+-----+-----+-----+-----+
| feedback_id | viewing_id | comments                                     | interest_level |
+-----+-----+-----+-----+
|          1  |          5 | wej kjklj fperjg;tkrhtlkrmrktlgj;ltrhkl | high          |
+-----+-----+-----+-----+
1 row in set (0.001 sec)

```

Relationship Between Tables

The system contains relationships between tables through foreign keys.

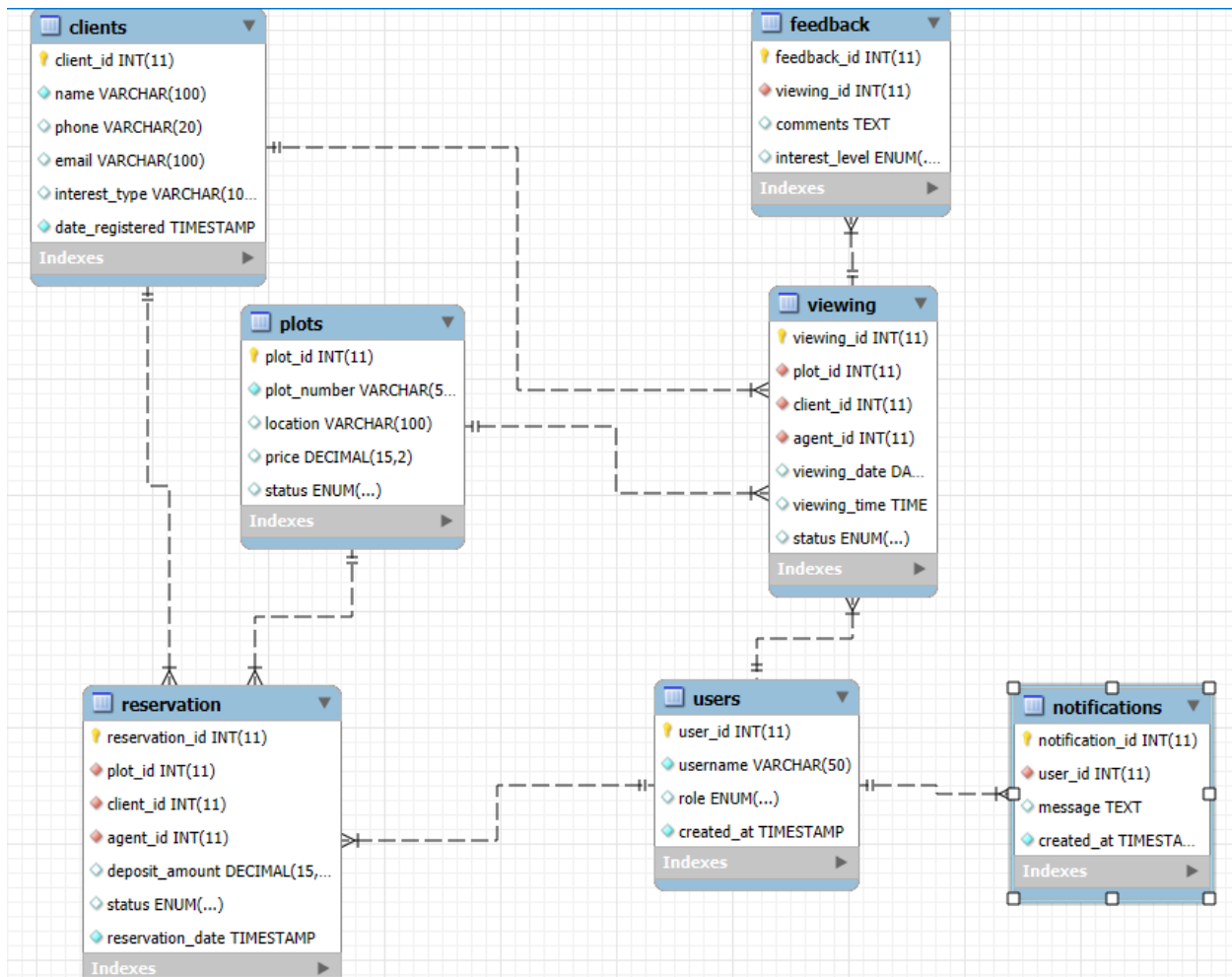
1. One user can schedule many viewings.
2. One client can schedule many viewings.
3. One plot can appear in many viewings.
4. One client can reserve many plots.
5. One plot can be reserved by clients.

Entity Relationship Explanation

The users table is connected to the viewing table through the agent_id field. The clients table is connected to both the viewing and reservation tables through the client_id field. The plots table is connected to the viewing and reservation tables through the plot_id field.

This relational structure ensures that all information in the system remains linked correctly.

ERD



Physical Design

The physical design describes how the system was implemented using actual technologies and database structures.

Implementation

Introduction

The implementation stage involved coding the web application, connecting it to the MySQL database, and testing the system functionalities.

Database Connection

The application uses a PHP PDO connection to securely communicate with the MySQL database.

```
<?php
```

```
$host = 'localhost';
```

```
$db = 'real_estate';
```

```
$user = 'root';
```

```
$pass = '';
```

```
try {
```

```
    $pdo = new PDO("mysql:host=$host;dbname=$db;charset=utf8mb4", $user, $pass, [
        PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
    ]);
```

```
} catch (PDOException $e) {
```

```
    die("Connection failed: " . $e->getMessage());
```

```
}
```

```
?>
```

User Interface

The user interface was designed using HTML and CSS to create a modern and attractive appearance.

The interface contains:

1. Client registration form

Register Client

Add Client

2. Viewing scheduling form

Schedule Viewing

-- Select Plot --

-- Select Client --

mm / dd / yyyy

Schedule

3. Plot reservation form

Reserve Plot

-- Select Available Plot --

-- Select Client --

Deposit Amount

Reserve

4. Plot display section

Available Plots

- Plot P001 - Kampala - Ntinda - UGX 25,000,000 - Status: reserved
- Plot P002 - Kampala - Kira - UGX 18,000,000 - Status: reserved
- Plot P003 - Wakiso - Entebbe Road - UGX 35,000,000 - Status: available
- Plot P004 - Mukono Town - UGX 12,000,000 - Status: available
- Plot P005 - Jinja - Bugembe - UGX 8,000,000 - Status: available

The system uses cards, colors, and spacing to improve user experience.

Future Improvements

The system can be improved in the future by adding:

1. User authentication and login system.
2. Report generation features.
3. Online payment integration.
4. Email and SMS notifications.
5. Search and filtering features.
6. Advanced dashboard analytics.
7. Mobile responsiveness.

These improvements would make the system more powerful and suitable for commercial use.

Conclusion

The Real Estate Management System successfully achieved its objectives by providing an automated solution for managing clients, plots, reservations, and property viewings. The project demonstrated the practical application of database management systems.

The system improved efficiency, reduced manual work, and ensured better organization of records. The use of relational database concepts, foreign keys, and ensured data consistency and reliability.

Overall, the project provided valuable experience in database design and system implementation.

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